

Kratzer on 'if'-clauses as restrictors

Topics in Language and mind

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1. True embedding versus merely orthographic embedding

Suppose we are theorising about the sentence

(1) Antony loves Cleopatra

trying to figure out what proposition it expresses, or what compositional semantic value to assign it. An important source of evidence concerns sentences which *embed* (1), such as

- (2) a. It is necessary that Antony loves Cleopatra.
- b. Either Antony loves Cleopatra or Cleopatra loves Antony.
- c. Caesar thinks that Antony loves Cleopatra.

We should expect that our compositional semantic value for (1) should fit together with compositional semantics for the other words in these sentence to yield correct semantic values for (2a–c). But our inquiry will go completely off the rails if we expect the same thing in the case of (3a–b):

- (3) a. Everyone who loves Antony loves Cleopatra.
- b. Antony loves Cleopatra more than anyone else.

'Antony loves Cleopatra' is not a *constituent* in (3a) and (3b). For example the constituency structure of (3a) seems to be [Everyone [who [loves Antony]]] [loves Cleopatra]; to work out the semantic value of (3a) we only need to know the semantic values of the nodes in this tree.

Similarly, (4) is not a constituent in (5a–b):

- (4) Dogs bark.
- (5) a. Some dogs bark.
- b. Most dogs bark.

The tree structure of (5b) is [Most dogs] bark, not Most [dogs bark].

Consider finally:

(6) Whenever I go outside, dogs bark

Although it is defensible to think that 'dogs bark' is a constituent in (6), it still seems implausible to think that its meaning can be put together by applying a 'Whenever I go outside' operator to the proposition expressed by (4) in isolation, which is roughly that barking is a characteristic behaviour of dogs. The idea of "generic" or "characteristic" behaviour doesn't seem to enter into the meaning of (6) at all. A possible diagnosis:

when uttered in isolation, (4) includes an unpronounced element GEN contributing this element of meaning: its logical form is something like '[GEN dogs] bark' or maybe 'GEN [dogs bark]'. But in (6), GEN does not appear at all.

2. Conditionals with adverbs of quantification

Consider:

- (7) If a man buys a horse, he pays cash for it.
(8) a. Sometimes, if a man buys a horse, he pays cash for it.
b. Always, if a man buys a horse, he pays cash for it.
c. Usually, if a man buys a horse, he pays cash for it.

Weak claim: the complete structure that would be contributed by (7) in isolation is not a constituent in (8a–c).

Strong claim: the string 'if a man buys a horse, he pays cash for it' does not correspond to any constituent in (8a–c). Their real compositional structure is one where 'sometimes' combines with one of the two clauses before they combine with each other—either [Sometimes [if a man buys a horse]] [he pays cash for it], or else maybe [If a man buys a horse] [sometimes [he pays cash for it]].

Evidence for weak claim: the proposition expressed by (7) has a 'generic' flavour which seems absent from, say, (8a).

Evidence for strong claim: the competing tree structures do a better job of predicting the intuitive meanings of (8a)–(8c).

- Strategy: take the 'if' clause and main clause to denote properties of events, and take adverbial quantifiers to denote functions from properties of events to functions from properties of events to propositions. For example: $\llbracket \text{sometimes} \rrbracket (F)(G)$ = the proposition that some events have both F and G; $\llbracket \text{usually} \rrbracket (F)(G)$ = the proposition that most events having F have G; etc.
- Given this semantic value, the AdvQ needs to combine semantically with the 'if'-clause and then the result needs to combine with the main clause.

3. Conditionals with modals

Kratzer favours a parallel account of sentences in which 'if'-clauses interact with modal verbs like 'must', 'can', 'should', 'would', 'have to', 'need to', 'be able to'; modal adverbs like 'probably', 'possibly', 'necessarily'; and modal adjectives like 'probable', 'likely', 'possible', 'necessary'.

Her favoured syntax seems to assign all of (9a–c) the logical form (9d):

- (9) a. Probably, if Yog had white, Zog won
b. If Yog had white, Zog probably won

- c. Probably, Zog won if Yog had white.
- d. [Probably [if Yog had white]] [Zog won]

Semantics for ‘probably’: in context, ‘probably’ denotes a function from propositions to functions from propositions to propositions—[[probably]](P)(Q) is true in those worlds where the relevant people’s evidence supports a high conditional credence for Q given P. (Or something like that.)

If this is right, then the truth conditions of (9a–c) don’t tell us anything about the proposition expressed by the bare conditional ‘If Yog had white, Zog won’.

4. Extending to attitude operators

In this paper Kratzer does not discuss sentences like (10a–b):

- (10) a. I am pretty confident that if Yog had white, Zog won.
- b. I am pretty confident that Zog won if Yog had white.

Up to now, we have been sentences like these as quite important data points for figuring out what’s expressed by ‘if Yog had white, Zog won’: we argued that the proposition in question must be one that it would make sense to be pretty confident in exactly when one’s conditional credence in Zog having won conditional on Yog having white was high. But this is all wrong if we adopt a view where the string ‘if Yog had white, Zog won’ doesn’t correspond to any constituent in (10).

5. Counterfactuals and their chances

For Kratzer the ‘would’ in counterfactuals behaves like any other modal: (11a) has an underlying structure like (11b).

- (11) a. If the coin were tossed, it would land heads
- b. [Would [if the coin were tossed]] [it land heads]

Since the ‘would’ is already there playing the role of the modal that combines directly with the ‘if’-clause, (12a) *does* have a constituent corresponding to (11a):

- (12) a. It is 50% likely that if the coin were tossed, it would land heads.

So, the Kratzer-style logical syntax does *not* disrupt the arguments we have been making earlier in the course where we used sentences like (12a) to draw conclusions about the propositions expressed by sentences like (11a).

6. Apparently bare conditionals

When there isn’t an overt (audible, visible) modal (or AdvQ) in a sentence with an ‘if’ clause in it, Kratzer posits that there is a covert (unpronounced) one. Sometimes this is an AdvQ along the lines of ‘always’ or ‘typically’ or ‘generically’, as in (7) (see above). But for standard indicative conditionals, Kratzer proposes that the hidden modal is an

epistemic necessity operator with a meaning similar to that of the overt modal ‘must’. For example, (13a) has the logical form (13b):

- (13) a. If the coin was tossed, it landed heads
 b. [MUST [if the coin was tossed]] [it landed heads]

‘MUST’ is context-sensitive like ‘must’: in our terms, it expresses a function that maps any proposition P to the function that maps any proposition Q to a proposition true in all worlds where the salient people’s evidence entails $P \supset Q$.

- Note that the rest of Kratzer’s story would work just as well no matter what she said about the semantics of this silent modal—she could even take it to be the material conditional! The choice of a ‘must’ like meaning is not fully motivated in the paper we read.
- Kratzer’s view allows her to say even when an overt modal is present, there are some cases where the covert modal is present too and is the one that interacts directly with the ‘if’ clause. So for example (14a) could perhaps have a LF like (14c) as well as one like (14b):

- (14) a. It’s possible that Jane was there if Mary was there
 b. [Possible [if Mary was there]] [Jane was there]
 c. [Possible $\emptyset$] [[MUST [if Mary was there]] [Jane was there]]

These would be predicted to have quite different meanings—(14b) will be equivalent to ‘it’s possible that Jane and Mary were both there’, while (14c) will be equivalent to ‘It’s possible that Mary couldn’t have been there without Jane’ (on an epistemic reading of the ‘couldn’t’).

7. Stacking ‘if’ clauses